

Contribution Information Sheet

Discover Artificial Intelligence — Springer Nature

A Controlled Reassessment of Support Vector Regression on the California Housing Benchmark with Feature Scaling, Feature Selection and Hyperparameter Tuning

Corresponding author	Emmanuel Adutwum
Affiliation	Soka University of America, Aliso Viejo, CA, USA
Email	eadutwum@soka.edu
Submission ID	6cf4e30e-1a5b-4ad2-99e5-2320f0737507
Stage	Major revision (resubmission)

1. Scientific question addressed

A recent comparative study (Preethi et al., *SN Computer Science*, 2025) reported that Support Vector Regression ranks last among the regression algorithms it compared on the California Housing benchmark. This paper asks a precise question about that specific result: *is the reported outcome a property of the algorithm, or of the configuration in which the algorithm was run?*

The question is one of evaluation methodology and reproducibility in applied AI. It is answered by reproduction rather than by assertion: the configuration described in the prior study is re-run directly, components are added to it one at a time, and the resulting attribution is tested for robustness to the order in which those components are introduced.

2. Methodology

The workflow is defined as an explicit nine-stage procedure (S1–S9) with stated inputs and outputs, illustrated as a pipeline diagram in the manuscript. Every stage that estimates anything from data — feature ranking, scaler assignment, scaler parameters, hyperparameters — is fitted on the training portion only and re-fitted from scratch inside every fold of every resampling scheme. The evaluation comprises:

1. **Direct reproduction** of the prior configuration (eight raw features, RBF kernel, a grid over C and ε only, no scaling), over 5 seeds.
2. **Equal-budget model comparison**: all 10 models receive identical features, identical training rows and an identical tuning budget.
3. **Corrected statistical inference**: 10 identical splits, with the Nadeau–Bengio corrected resampled t -test as the primary paired test and the confidence interval computed correctly.
4. **Order-independent ablation**: a four-factor full factorial design evaluated in all 32 cells, with Shapley attribution over all 24 orderings.
5. **Validation under dependence**: spatially blocked cross-validation using k -means blocks on standardised centroids, plus feature selection repeated inside every fold and fully nested estimates.
6. **External replication**: the workflow applied unchanged to King County house sales and Ames Housing.

3. Key findings

1. **A correction to the record.** The value of $R^2 \approx 0.60$ widely associated with the prior study’s SVR is in fact the score of its linear, ridge and polynomial-ridge models. Its SVR-RBF result is $R^2 \approx 0.15$. The configuration it describes reproduces here at -0.020 with library defaults and 0.514 when tuned, bracketing that value; adding a single standardisation step to the identical pipeline yields 0.766 .
2. **Order-independent attribution.** The apparent contribution of scaling varies by 0.7441 in R^2 depending only on where it is placed in the ablation sequence. The order-independent Shapley values are 0.2723 (scaling), 0.1882 (tuning), 0.1789 (selection) and 0.1505 (derived features). Single-path ablation increments are therefore upper bounds, not independent contributions.
3. **Ranking under equal conditions.** With identical data and an identical tuning budget the tuned SVR ranks 4 of 10, behind gradient-boosted trees, and the paired differences survive the Nadeau–Bengio correction.
4. **Spatial optimism dominates.** Spatially blocked cross-validation lowers the estimate by 0.5180 relative to random partitioning — a larger effect than any pipeline component measured in the study.
5. **Three negative results.** The feature-specific scaling strategy proposed in the earlier version of this work does *not* outperform plain standardisation (0.6925 against 0.7453 on an identical split); the distribution-aware scaler rule introduced to replace it does *not* transfer, preserving the component ordering on 1 of 3 datasets against 3 for plain standardisation and carrying a negative Shapley value on Ames (-0.1558); and the derived features contribute little once selection and tuning are present. All three are reported rather than omitted.

4. Specific contributions to the AI field

The contribution is methodological and diagnostic. *No new learning algorithm is proposed*, and the paper makes no claim to have discovered that kernel methods require feature scaling — a result established decades ago and cited as such.

1. **An order-independent ablation procedure for preprocessing pipelines.** Evaluating the full factorial design and attributing components by Shapley value removes the dependence of reported increments on the sequence in which components are introduced. This is transferable to any staged pipeline and is what revealed that the earlier single-path attribution was an artefact.
2. **A worked correction of a published comparison.** A documented case in which a benchmarking result is traced to configuration rather than to algorithm, including the correction of a misreading in the earlier version of this work.
3. **An evaluation protocol for dependent tabular data.** Corrected confidence intervals, paired testing on identical splits under the Nadeau–Bengio correction, selection inside the resampling loop, and spatially blocked validation — with the magnitude of each source of optimism quantified rather than asserted.
4. **Fully generated reporting.** Every experiment writes a JSON record, and every table, figure and in-text number in the manuscript is generated from those records. No value is typed by hand, so text, tables and figures cannot diverge.

5. Significance and fit for Discover Artificial Intelligence

The journal’s scope covers machine and deep learning, data analytics and AI-enabled data-driven techniques. Beyond the housing application, the paper addresses a question central to applied AI: how much of a reported algorithm ranking is a property of the algorithms and how much is a property of the experimental pipeline around them. The answer given here is specific and measured — and, importantly, includes the finding that the validation design matters more than any preprocessing choice tested.

Conclusions are explicitly restricted to the datasets and configurations examined. The California Housing dataset ships with scikit-learn, and both additional datasets are on OpenML, so every result is reproducible by any reader with a Python environment.

6. Intended audience

- Researchers designing or reviewing comparative benchmarking studies
- ML practitioners evaluating kernel methods for tabular regression
- AutoML and pipeline-design researchers, for whom preprocessing sensitivity and order-dependent attribution are core concerns
- Anyone modelling spatially structured tabular data, given the spatial-validation result

7. Reproducibility

All code, the JSON record of every experiment reported, and the scripts that generate every table and figure are available at <https://github.com/emmanueladutwum123/svr-california-housing>.

<code>main.tex</code>	Manuscript source (Springer Nature <code>sn-jnl</code>)
<code>common.py</code>	Shared data loading, scaling, selection and pipeline
<code>e1--e8*.py</code>	One script per experiment, each writing a JSON record
<code>make_numbers.py, make_tables.py, make_figures.py</code>	Generate every number, table and figure from those records
<code>results/*.json</code>	Complete result records for every configuration reported

Datasets: `sklearn.datasets.fetch_california_housing`; OpenML `house_sales` (ID 42731); OpenML Ames Housing (ID 42165). All public, no special access required.

8. Keywords

Support Vector Regression; feature scaling; ablation analysis; Shapley attribution; reproducibility; spatial cross-validation; nested cross-validation; paired significance testing; benchmarking methodology; California Housing dataset.

9. Conflict of interest

The author declares no competing interests.

10. Data availability

All datasets analysed are publicly available and no new data were generated. The California Housing dataset is distributed with scikit-learn and archived at StatLib (<http://lib.stat.berkeley.edu>).

cmu.edu/datasets/houses.zip); King County house sales is at <https://www.openml.org/d/42731>; Ames Housing is at <https://www.openml.org/d/42165>.

11. Use of generative artificial intelligence

The author used a large language model (Anthropic Claude) as an assistive tool for drafting and copy-editing prose, for implementing and refactoring the experiment scripts, and for generating tables and figures from the result records. The research questions, experimental design, interpretation of results and all scientific claims are the author's own. The author has reviewed and verified every number, statement and reference, and takes full responsibility for the content.